Frank Shan

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Highlights

- Backend-leaning builder experienced in **distributed system design**, data modeling, and performance optimization, with a focus on correctness, scalability, and maintainability.

Education

University of Waterloo | *CGPA: 92.3/100* — *Sep 2024 - Apr 2029*
Bachelor of Computer Science, Honours Computer Science, AI Specialization

Experience

Software Developer Intern | *Richmond, BC* — *Sep 2025 – Apr 2026*
Ziyutec Inc.

- Designed a modular **service-oriented architecture** (NestJS + PostgreSQL + Redis), enabling independent feature evolution, fault isolation, and scalable backend growth.
- Built a **semantic search system** (OpenSearch) with hybrid lexical + vector retrieval and ranking, enabling automated resource discovery and recommendation workflows.
- Built a **RAG system** indexing **1,000+ enterprise documents**, improving retrieval **faithfulness and contextual relevance by 16%** (DeepEval) via hybrid search, reranking, and context-aware chunking.
- Developed an internal **workflow automation agent** integrating database queries and backend tools, reducing manual operational workload by **20%** across a **50-person** team.

Research Assistant | *Waterloo, ON* — *Feb 2026 – Present*
University of Waterloo

- Designed experiments for a probabilistic programming system (PAL) compiling symbolic constraints into tensor operations, enabling **neuro-symbolic training** with strict structural guarantees.
- Built a hybrid NER pipeline combining frozen language model embeddings with constraint-based decoding, improving **prediction validity** and early-stage training stability.

Projects

Mini VLM | *Python, PyTorch, SigLIP-2, Qwen3-0.6B* — github

- Built a LLaVA-style **Vision-Language Model**, integrating **SigLIP-2** with **Qwen3-0.6B** via a modality projector, achieving **32.9% zero-shot MMStar accuracy** under a **¡1B parameter** constraint.
- Designed a high-throughput multimodal training pipeline with **dynamic batching** and token-aware scheduling, maximizing GPU utilization on **16GB VRAM hardware**.

Mini VLLM | *Python, PyTorch* — github

- Built a lightweight **LLM inference engine** with continuous batching, request scheduling, and pluggable **paged KV-cache** architecture for efficient memory management.
- Developed a ShareGPT-based benchmarking suite, achieving up to **59% higher throughput** and reduced time-to-first-token (TTFT) with paged attention.

Resume Builder | *Python, Jinja2, prompt_toolkit, LaTeX, LangGraph* — github

- Built a **multi-step resume generation agent** (LangGraph) that maps experience to job descriptions via structured reasoning and template-driven generation.

Skills

Programming Languages: Python, TypeScript, JavaScript, C++, SQL, HTML, CSS
Frameworks: NestJS, Next.js, React, Express, FastAPI, LangGraph, PyTorch
Tools: AWS, PostgreSQL, Linux, Docker, Git

Professional Certificates

Stanford Online — *December 2025*
XCS224N Natural Language Understanding — credential